

Biofiltration System Design to Remove Odor Causing Compounds and VOC's

BCHE 3420

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SUMMARY

A biofilter is a type of filter that contains a biological media in which a fluid is sent through. In the media, microbes are present that consume unwanted chemicals, releasing more environmentally friendly ones. Biofilters can be configured with a plethora of materials and microorganisms, all with unique parameters affecting the filtration process. This type of filtration is generally more effective than systems widely used today and offers a promising solution to industry problems with volatile-organic-compounds and odors.

This report designed an up-scaled biofilter based off of the data collected experimentally from a bench-scale biofilter. Both biofilters had a cylindrical design and were packed with BIOREM BIOSORBENS® media.

Hexanal was determined to be the rate-limiting aldehyde for the tasked biofiltration process when compared to 3-methyl butanal and 2-methyl butanal. The compound was determined to follow first-order kinetics, of which the up-scaled biofilter was based on.

1 INTRODUCTION

1.1 Air Pollution Issues from Industrial Operations in the U.S.

One of the biggest problems humanity has caused is air pollution. This pollution is caused by the many chemicals dispersed from industrial operations, negatively affecting the environment. These pollutants affect the whole ecosystem as they are absorbed from the air into the water and soil and from all three of these media, they enter the bodies of every living thing. From the air we breathe to the food we eat, these chemicals enter our bodies in every way possible leading to a plethora of health problems.

There are tens of thousands of different organic compounds in the air affecting living organisms and the atmosphere. These organic compounds are termed ‘volatile organic compounds’ (VOCs), being defined as organic compounds having a vapor pressure greater than 10 atm at 25°C, a boiling point of up to 260°C at atmospheric pressure, and 15 or fewer carbon atoms [1]. The primary contribution of human VOC emissions is fossil fuel use [1], with the industrial and electrical power sectors using contributing the most [2]. These VOCs can cause eye, nose, and throat irritation, central nervous system damage, and some are known to cause cancer in humans and animals [3].

In addition to these VOCs, there are also odor-causing chemicals such as sulfur dioxide (SO_2), which is currently the only regulated air pollutant with a strong, pungent odor [4]. These odors, while most are not nationally regulated due to low toxicity, are a problem to industry due to complaints as they are irritating to the citizens living nearby. Of course, these chemicals smell bad, but they can also cause symptoms such as headaches, nasal congestion, depression, and heart palpitations depending on how sensitive someone is to the particular odor [5].

With a rise in urbanization and odor complaints, a need for low-cost, effective treatment options has surfaced to minimize industrial VOC and odor emissions. Biofilters, which are filters with a biologically active material that degrade unwanted, biodegradable toxins from a fluid, offer a promising solution to this problem, [6].

1.2 Biofiltration History:

The idea of treating odorous gasses by biological methods can be found earliest in 1923 when H. Bach discussed the control of H_2S emissions from sewage treatment plants. Reports on the application of this from the 1950s were published in the U.S. and West Germany, with Pomeroy receiving U.S. Patent No. 2,793,096 in 1957 for the concept of using soil as a biofilter. Nuremberg, West Germany implemented a soil bed to control odors from an incoming sewer main around 1959 [6].

The first research in the U.S. on biofiltration was conducted by Carlson and Leiser in the 1960s, in which they successfully installed several soil filters at a wastewater treatment plant. This research showed how instead of sorption, biodegradation accounted for odor removal [6].

After Carlson and Leiser's research, several other researchers in the U.S. further studied the concept of a soil bed filter. Much of the knowledge today is credited to Hinrich Bohn, who investigated this technology for more than 15 years [6]. In 1985, Bohn concluded that "the soil bed is at least equivalent and probably superior to other known odor control methods, including incineration and wet scrubbing," while being more cost-effective than the technology used at the time [7].

While soil beds are cheap and have been shown to effectively eliminate certain odors and VOCs, their use is limited by the low biodegradation capacity of soils and a large amount of space required for such a filter. With these problems, it was estimated that less than 50 soil beds were installed in the U.S. and Canada by the early 1990s, along with little research published. The small amount of research published inferred little attention to the technology in these countries.

However, in the early 1980s, biofiltration use increased in Germany in industrial facilities due to new federal regulations for the control of VOCs and a well-funded development program run by the West German Federal Environmental Agency, Umweltbundesamt, as well as through engineering firms [6]. New systems were developed primarily in Germany and the Netherlands, redesigned with new air-distribution systems and structural support media such as bark, wood chips, and polystyrene balls, reducing filter bed compaction and distributing airflow evenly [8].

In the 1980s and 1990s, progression in biofiltration increased rapidly through Europe while remaining slow in North America, with the development of computer-operated enclosed systems. Inorganic filter bed media were also developed, with examples including granular activated carbon, powder-activated carbon-coated polystyrene, and ceramics. These new materials improved the porosity and life expectancy of the filter beds [8].

Today, with an increasing awareness of the seemingly infinite amount of pollutants in the environment, biofilter research throughout the world continues in an effort to clean up these pollutants. With an increased effort to save the environment from deposits of pharmaceuticals, personal care products, and microplastics, biofiltration research is constantly improving [9, 10].

1.3 How do biofilters work?

A biofilter is a layer of organic material that supports a microbial population. This organic material is typically a mixture of compost and wood chips, though new materials are constantly being researched. Air containing unwanted chemicals is then forced through this material of which microbes consume and produce carbon dioxide and water. When constructing a biofilter, key factors include: gas residence time (amount of time the gas spend in the filter), moisture content of the filter material, sizing of the biofilter bed, fans that push the gas through the filter, the biofilter media, moisture control, operation and management, and of course, cost of construction and operation [11].

There are two configurations of biofilters:

1. *Open beds* are the most prevalent configuration, which are typically 10-18 inches deep and much larger than closed bed biofilters. These types of biofilters are typically built outdoors to be exposed to weather like rain, snow and temperature extremes [11].
2. *Closed beds* are mostly enclosed while having an exhaust port for venting the filtered air. These types of biofilters usually treat smaller airflows and have a deeper media of around 2-3 feet or more when compared to open bed variations to reduce space required for treatment. These types are also generally more expensive [11].

For all biofilter configurations, the air containing unwanted chemicals is exhausted by a fan which is then uniformly distributed through the biofilter media. Microorganisms attached to the media create a biofilm in which the microbes oxidize biodegradable gases. This oxidation creates carbon dioxide, water, mineral salts, and biomass. After, the cleaned exhaust air leaves the filter[11].

To design a biofilter, one must determine the volumetric flow rate of air to be treated, specific air contaminants and concentrations, media characteristics, biofilter area constraints, moisture control, maintenance, and cost [11].

Volumetric flow rate must be determined for the application of the biofilter, eg. for a biofilter used to treat ventilated air exhausted from a livestock building, the maximum ventilation rate of the building would be used [11].

The media used in the biofilter must provide a suitable environment for microbial growth while maintaining a high porosity in order to allow for air to flow undisturbed. Moisture holding capacity, nutrient content, life expectancy, and cost should also be considered when determining media [11].

As biofilters use microorganisms to degrade unwanted chemicals from an air flow, the number and type of microorganisms is important. When using a natural media such as soil or compost, microorganisms are naturally occurring and may be used in the filter. However, these microbes may need a necessary conditioning period for adaptation of the chemicals to be degraded, in turn limiting the biofilter efficiency during this period [11].

1.4 Comparison to other filtration technologies:

Biofiltration has several advantages compared to traditional air pollution control (APC) technologies including: low capital and operating costs, low energy requirements, no additional chemicals or fuel requirements, minimal maintenance requirements, the absence of residual products requiring other treatment or disposal, and above all, public acceptance as a “natural” process. These types of filters can also readily deal with multiple contaminants due to the nature of microbes [12].

APC technologies can be broadly grouped as thermal or catalytic incineration, physical or chemical treatment, and biological degradation. While incineration effectively destroys the organic contaminants, large amounts of fuel are needed, in turn possibly producing undesirable byproducts. While there are ways to reduce the amount of fuel needed for such a system, these ways can be offset through initial and maintenance expenses. Chemical oxidation is typically ineffective for hydrocarbons and other slow reacting compounds, with the added production of wastewater that then needs to be treated [12].

Using adsorption on activated carbon is not as effective as a biofilter as the carbon is costly and once saturated, becomes a hazard that needs to be either regenerated or transported to a landfill. While these process have the benefit of being able to treat multiple pollutants at high concentrations, they fail to treat such at low concentrations [12].

Biofiltration utilizes microbial reactions at normal temperatures and pressures, making such a technology safe and energy saving- as there is no need to control such parameters in comparison to other filtration methods. However, biofiltration systems require longer residence times of up to 60 seconds, of which can inhibit their use to specific sites [12].

Costs of biofiltration varies depending on the biofilter set up with maintenance generally costing up to 20% less than that of other technologies. Without the production of other pollutants, even more money is saved as there is no added cost for treating those [12].

2 PROBLEM DEFINITION & PROCESS DESCRIPTION

The purpose of this report was to determine the design of a scaled-up bench-scale biofilter (Fig. 1) with the intent of removing concentrations of VOCs and odor-causing chemicals from the gaseous waste of an industrial process. This biofilter was instructed to use BIOREM's BIOSORBENS ® media [13]. The up-scaled biofilter was adjusted for a 95% degradation of a determined rate-limiting aldehyde, which was measured through experimentation of the original bench-scale bioreactor. The experimental concentrations can be found in Appendix A, of which the concentrations of the chemicals were measured along the length of the bench-scale bioreactor. These concentrations identified the kinetic parameters, allowing for determination of the rate-limiting compound. Additional parameters for the scale-up included composition of the industrial waste influent (which was given and included in Table 2 in Appendix A) and residence time. The residence time was recorded as empty bed contact time (EBCT), which is the amount of time that a unit volume of the influent remains within an empty biofilter. The EBCT gives a predicted residence time that is greater than the actual and with this overestimation, ensured a degradation of at least 95%.

When determining the geometry of the biofilter, Mike Harman of BIOREM was contacted to give a ratio between the cross-sectional area and length to ensure structural stability, of which it was concluded that the length should not exceed three-times the cross-sectional diameter. Harman recommended using concrete to surround the media, and it was also concluded that the final design should contain an empty volume of space before and after the media to allow for adequate air flow. In the final design, this extra space was not

determined as the report was tasked to scale up the original bench-scale biofilter of which no empty space was noted. Furthermore, it was noted by Harman that for the reported inflow volumetric flowrate for the final design, the biofilter should have a rectangular-prism geometry. However, a cylindrical geometry was kept to keep similar parameters of the original bench-scale biofilter design.

Looking at the schematic for the original bench-scale biofilter (Fig 1), the waste airflow feed was inputted through the filter with a flow controller. A flow meter was attached in order to measure the volumetric flow rate, which can be used to make sure the controller was functional. The air was humidified to avoid a drying out of the media, heated to maintain isothermal conditions, and in the mixer, mixed with nutrients supplied gradually through a syringe pump. These three steps ensured an environment crucial to the livelihood of the microbes carrying out the filtration.

To experimentally determine the kinetics of the chemicals in waste gas, five ports (probes) were placed on the reactor, measuring the concentrations of select chemicals as they traveled through the reactor. These ports were indicated along the length of the reactor: **port 1** (the inlet) at 0m (0% of the length), **port 2** at 0.15m (30% of the length), **port 3** at 0.25m (50% of the length), **port 4** at 0.35m (70% of the length), and **port five** (the outlet) at 0.5m (100% of the length). For the final design of the scaled-up biofilter, these ports were to be placed at the same length percent to keep similar design to the original bench-scale biofilter. The treated air was then expelled through a fume hood.

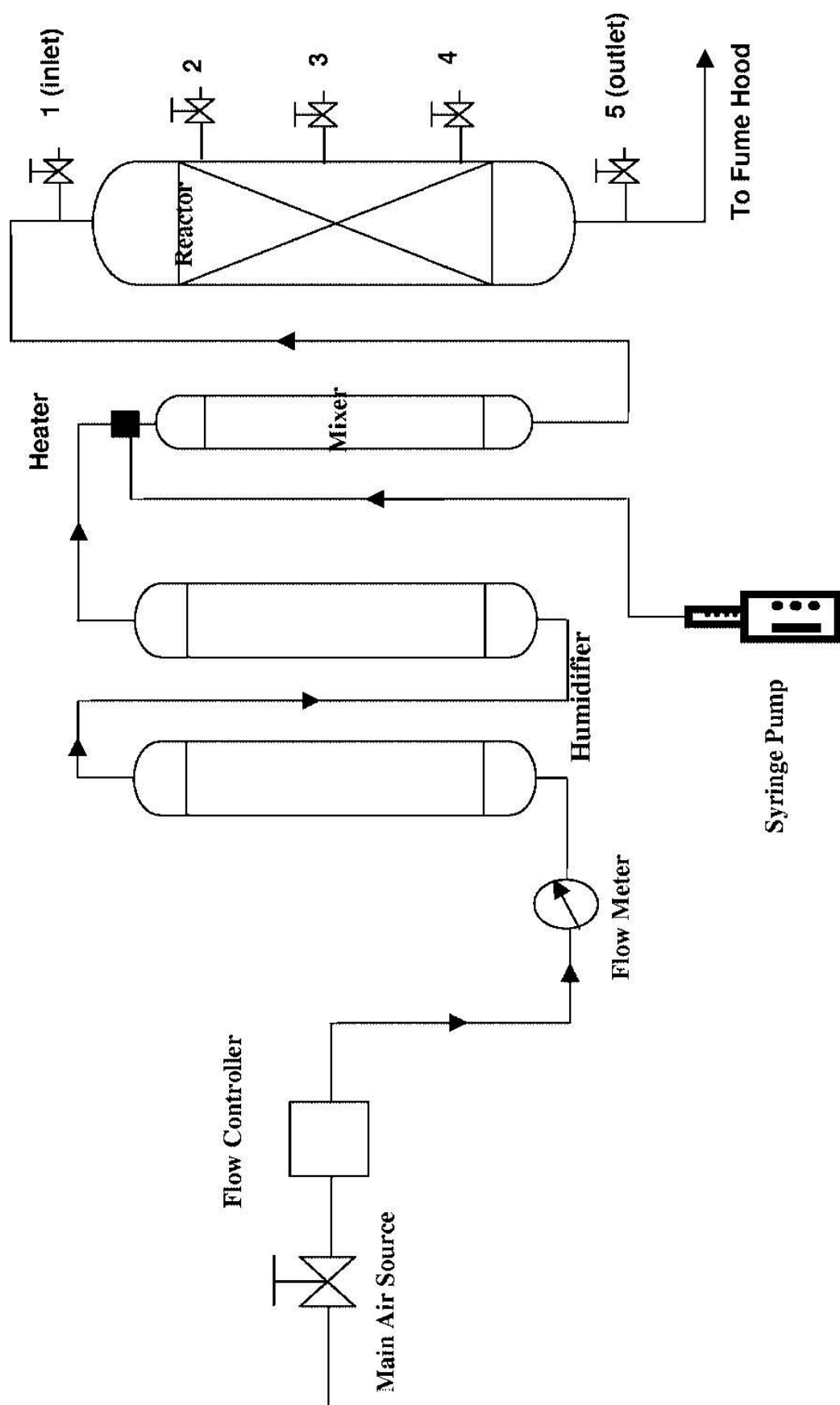


Figure 1 Bench-scale biofilter process flow diagram [13].

3 RESULTS & ANALYSIS

When designing the biofilter, the bench-scale biofilter concentration data (c_A = concentration, c_{A0} = initial concentration) for 5 chemicals (methanethiol CH_3SH , 3-methyl butanal 3-MB, 2-methyl butanal 2-MB, hexanal, and dimethyldisulfide DMDS) were given (found in Appendix A), of which the rate-limiting aldehyde (3-MB, 2-MB, or hexanal) was determined to be *hexanal*. This was achieved by comparing concentration rate equations for zeroth, first, and second order. Each equation gave a function that could be written as $y = mx + b$, of which y was plotted against x and m = rate constant (k), shown in Figure 2.¹ Then, linear regression was used to determine the slope and R^2 of each plot. The R^2 values for each aldehyde were compared, with the highest determining whether the specific aldehyde followed *zeroth*, *first*, or *second-order* kinetics.

R^2 is the determining statistic for rate order as R^2 quantifies how much variation of a dependent variable is explained by the independent variable. The higher the R^2 , the more the independent variable can explain the dependent variable. For each aldehyde, the R^2 's were determined for each model describing the reaction order. Figure 2 shows that 3-MB and 2-MB were both best modeled by first-order kinetics due to their high R^2 -values. However, the reaction order for hexanal could not be confidently described by this method, as first, there were too few data points for the first order plot, and second, the R^2 -values for first and zeroth order were very similar. To account for this, both first and zeroth order kinetics were considered for hexanal. None of the aldehydes were found to be of second-order.

After each aldehyde's order was determined, the slope of the respected plot gave the rate constant k , allowing for determination of the rate-limiting aldehyde. The rate-limiting aldehyde was found by finding which one had the lowest rate, which depended on reaction order. It was concluded that hexanal was the rate-limiting aldehyde, regardless if it had zeroth or first order kinetics, as if zeroth order, rate would equal k , which is smaller than 3-MB and 2-MB's rates. If hexanal had first order kinetics, its rate would equal $k \cdot c_A$, and since both k and c_A for hexanal are smaller than for the other two aldehydes, its rate must be as well.

The volume (V) for the up-scaled biofilter with a 95% degradation rate and volumetric flow rate of $Q = 100,000 \text{ ft}^3/\text{min}$ ($= 2831.68 \text{ m}^3/\text{min}$) was then found with the calculated kinetics of the rate limiting aldehyde. The composition of the air to be filtered in the new up-scaled reactor can be found in Table 2 in Appendix A. Considering hexanal had yet to have its reaction order determined, two volumes were calculated- one for each type of kinetics

¹The y -variables differed for each order (which are labeled on each plot), while the x -variables were always time (min).

considered.

Considering zeroth-order kinetics:

$$\begin{aligned}
c_A &= c_{A0} - kt \\
&= c_{A0} - k \frac{V}{Q} \\
0.1 &= 2 - 10.332 \frac{V}{2831.68} \\
V &\approx 521 \text{ m}^3
\end{aligned} \tag{1}$$

Considering first-order kinetics:

$$\begin{aligned}
\ln(c_{A0}/c_A) &= kt \\
&= k \frac{V}{Q} \\
\ln(2/.1) &= 2.4259 \frac{V}{2831.68} \\
V &\approx 3500 \text{ m}^3
\end{aligned} \tag{2}$$

Since the V was larger when considering hexanal having first-order kinetics, that V was used. This was because, if hexanal was actually zeroth-order, the larger V would be able to provide proper degradation regardless. With Mike Harman's (BIOREM) recommendation of making the biofilter have no more than a 3:1 diameter-to-length ratio ($D:l$), the diameter for the biofilter was found to be:

$$\begin{aligned}
3D &< l \\
3500 &= \pi(D^2)/4 \cdot l \\
3500 &= \pi(D^2)/4 \cdot 3D \\
D &= 11.41m \\
l &= 34.23m
\end{aligned} \tag{3}$$

Since the biofilter was designed with concrete surrounding the inner media, no consideration was put in place in trying to limit the surface-area-to-volume ratio, as concrete is very cheap in comparison to the price of the media, and the amount of the media used is equal to the volume calculated.

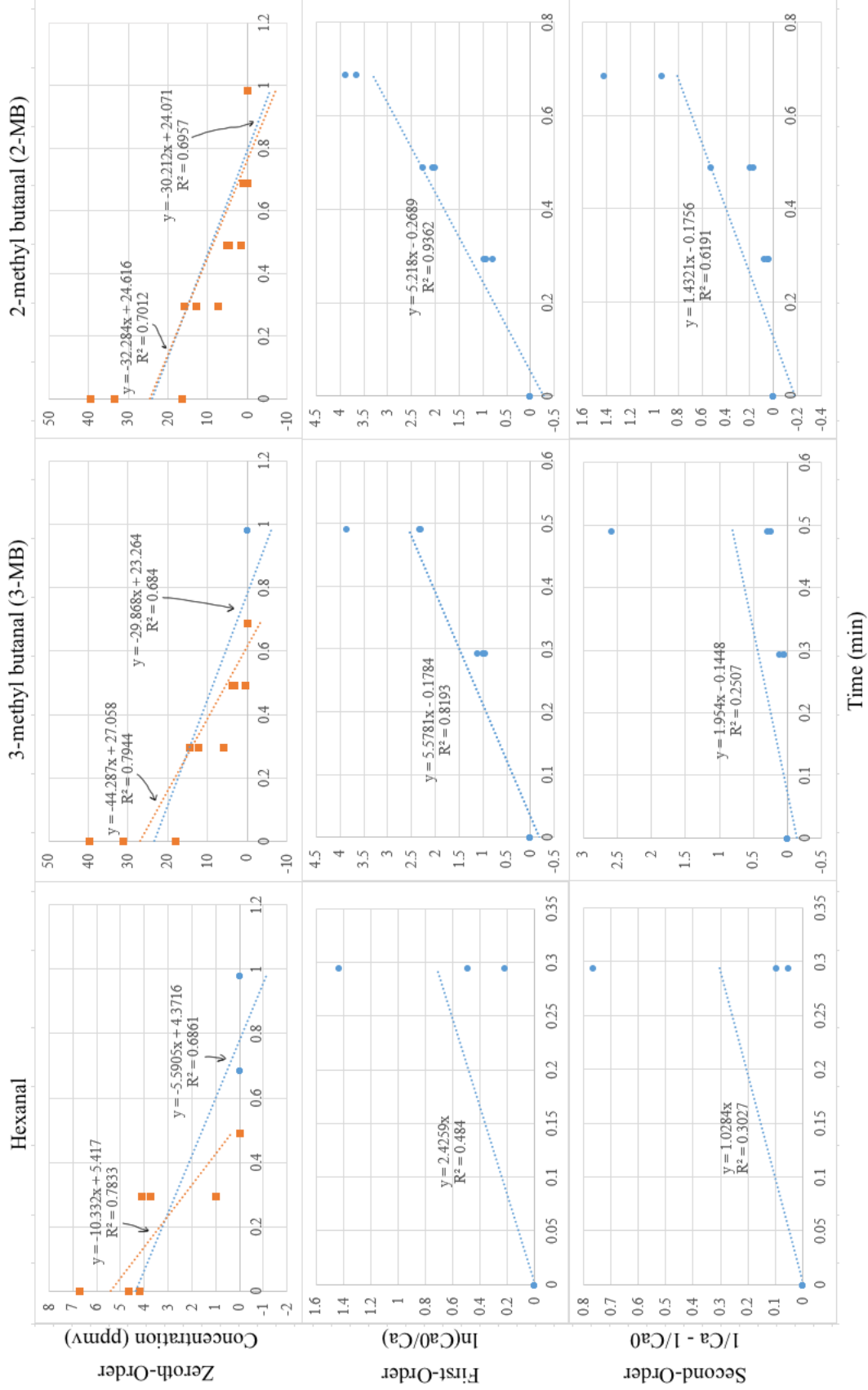


Figure 2 Determination of kinetics for the three aldehydes, hexanal, 3-methyl butanal, and 2-methyl butanal. Equations used: zeroth-order: $c_a = c_{a0} - kt$, first-order: $\ln(c_{a0}/c_a) = kt$, and second-order: $1/c_a - 1/c_{a0} = kt$. t was determined by using the reactor volume at a given length divided by the volumetric flowrate. Note, for zeroth-order, two data sets were used. They both contain the same data, except the line denoted by the orange ■ does not contain multiple 0-values for concentration. The orange ■ data was ultimately used for calculations.

After the volume was calculated, the concentration of hexanal (c_A) given the position in the reactor (denoted as the length, l) was plotted in PolyMath software, shown in Figure 3. The equation used to determine $c = f(l)$ was determined by Equation 3. The derivation of this equation can be found in Equation 8 in Appendix B.

$$c_A = c_{A0} \cdot e^{-\theta \cdot l} \quad \text{where} \quad \theta = k \cdot \pi(D^2)/4 \cdot Q^{-1} \quad (4)$$

Shown in the figure is how at the maximum length of the biofilter (34.23m), the concentration reached 0.1 ppmv, displaying how the designed biofilter successfully degrades hexanal to 95%.

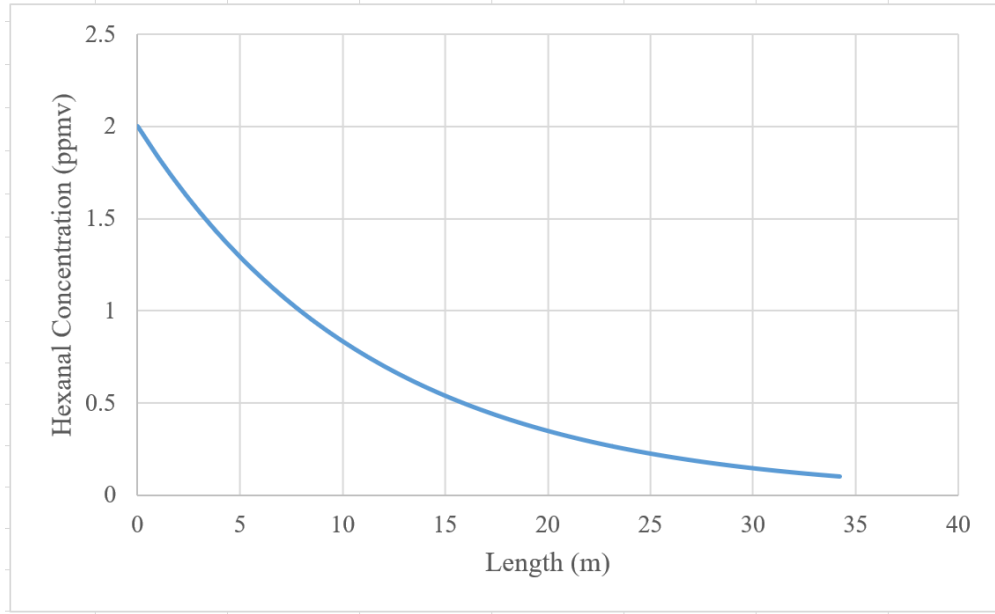


Figure 3 Hexanal concentration (c_A as a function of biofilter length, l)

3.1 Key Assumptions:

Since the initial concentration of hexanal to be filtered is very small (making c_A at the inlet $\approx$ outlet c_A), pressure drop and change in volumetric flowrate were not considered. This is shown by:

For pressure (P) drop:

$$\begin{aligned} P &= nRT/V \\ n_{inlet} &\approx n_{outlet} \\ P_{inlet} \approx P_{outlet} &= nRT/V \end{aligned} \quad (5)$$

For change in volumetric flow rate (Q):

$$Q = Q_0 \left[\frac{F_T}{F_{T0}} \right] \left[\frac{P_0}{P} \right] \left[\frac{T}{T_0} \right]; (z = z_o) \quad (6)$$

It should also be noted that for a biofilter, once set up, it will not be as effective for some time due to the microbes adjusting to the composition of the air flowing through the media.

Also, when developing a bioreactor, the experimental data displayed how DMDS increased. This increase is due to anaerobic degradation from the microbes and with a longer biofilter, less oxygen will make it down the length, leading to an increase in DMDS formation [14]. A solution would be to pump oxygen through longer biofilter designs.

4 CONCLUSIONS & RECOMMENDATION

Hexanal was determined to be the rate-limiting aldehyde for the tasked biofiltration process when compared to 3-methyl butanal and 2-methyl butanal. The compound was determined to follow first-order kinetics, of which the up-scaled biofilter was based on. In order to achieve a 95% degradation of hexanal, the biofilter composed of BIOREM's BIOSORBENS ® media and concrete exterior was designed with cylindrical geometry, with the diameter equal to 11.41m and the length equal to 34.23m. To determine the length and diameter given the volume, it was recommended that the length should not exceed three-times the volume in order to keep the system structurally sound.

Before construction of such system, it is recommended that the bench-scale biofilter be tested with air of the same composition of the air to be treated. This is because the experimentally tested air had a different composition than the air that will be filtered, of which a different chemical could be rate limiting. If that turns out to be the case, the effectiveness of the biofilter will be less than determined, leading to possible regulatory concern.

It is also recommended that the experimental bench-scale bioreactor have more ports determining chemical concentration. This is because when determining chemical kinetics of the aldehydes, many of the plots had few data points to be modeled. With more data to be analyzed, better relationships can be concluded from the data model. The data given provided evidence that hexanal was the rate-limiting aldehyde, yet there was a small amount of data to conclude so. This is shown through the small R^2 values for every plot involving hexanal. With more data, this whole biofilter design could easily be determined inefficient.

With more ports, there would also be more information given about the rate of degradation for each measured chemical. With the current data, a lot of the chemicals measured only had a few measurements before their concentrations reached zero, making it difficult to accurately analyze their degradation rates as the rate is constantly changing with concentration.

Another recommendation is to include an empty gap before and after the media in the biofilter. This was recommended by Mark Harman of BIOREM, indicating that with such gaps, the air would have time to "spread out" before reaching the media, allowing for greater media effectiveness. Without the gaps, the air would not fully come into contact with the whole area of the media near the beginning. This concept is displayed in Figure 4. As the media is expensive, these gaps will save costs on the design of the biofilter.

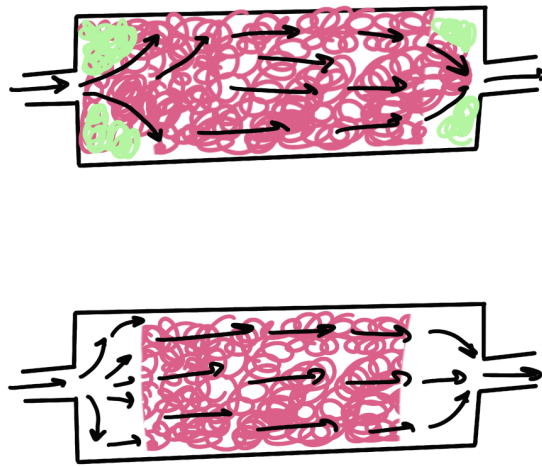


Figure 4 Biofilter designs comparing a design with and without empty gaps before and after the media. The arrows represent air flow, and the green represents parts of the media not in contact with air.

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A RAW DATA

Table 1 Biofilter Kinetic Data for Degradation of Hexanal, 2-methylbutanal, and 3-methylbutanal (Q=4 L/min, D=0.1m, total synthetic matrix packing length of 0.5 m). Concentrations are in ppmV. Note port 1 is 0.15 m, port 2 is 0.25 m, and port 3 is 0.35 m, all from the inlet. 3-MB is 3-methylbutanal, 2-MB is 2-methylbutanal, and DMDS is dimethyldisulfide [13].

Compound→	CH ₃ SH	3-MB	2-MB	Hexanal	DMDS
Reactor Position, m ↓					
<u>Date: 12/1/06</u>					
Inlet	16.05	39.64	39.6	6.7	1.68
Port 1	15.10	14.55	15.81	4.1	1.97
Port 2	14.57	3.87	5.14	0	2.03
Port 3	11.25	0	1.04	0	2.2
Outlet	9.11	0	0	0	2.3
<u>Date: 12/4/06</u>					
Inlet	20.03	31.34	33.55	4.66	1.62
Port 1	16.36	12.15	12.8	3.75	1.72
Port 2	13.61	3.15	4.53	0	1.95
Port 3	11.68	0	0.69	0	2.09
Outlet	8.32	0	0	0	2.2
<u>Date: 12/11/06</u>					
Inlet	21.0	18.0	16.36	4.2	0
Port 1	17.69	5.96	7.39	1	1.73
Port 2	14.34	0.38	1.7	0	2.00
Port 3	11.55	0	0	0	2.00
Outlet	8.77	0	0	0	2.16

Table 2 VOC concentrations in gas stream to be treated [13].

Compound Identified	Concentration, ppmv	
	Sample 1	2
Methanethiol	22.8	19.6
Propanal	8.8	6.4
2-Methylpropanal (Isobutanal)	11.1	14.1
3-Methylbutanal	17.1	12.0
2-Methylbutanal	27.2	30.3
Hexanal	3.2	0.8
Octane	0.01	0.04
2-Heptanone	P	ND
Heptanal	P	ND
Dimethyl disulfide	0.3	0.3
Benzene	ND	P
Toluene	P	ND
Total (ppmv)	90.51	83.54
P, Present but not quantified		
ND, Not detected		

Table 3 Physical properties of biocatalysts [13].

Properties	BIOSORBENS Media	Compost
Bulk density, dry basis	1447 kg/m ³	730 kg/m ³
Water-holding capacity	25% (dry basis)	50–60% (dry basis)
Particle size	5–25 mm	2–10 mm
Surface area	47.6 m ² /g	3.6 m ² /g
Life expectancy	Permanent	5 yr

B DERIVATIONS

Integral method for determining rate constant k for first-order:

$$\begin{aligned} -r_A &= k \cdot c_A \\ \frac{dc_A}{dt} &= k \cdot c_A \\ \frac{1}{c_A} dc_A &= k dt \\ \int_{c_{A0}}^{c_A} \frac{1}{c_A} dc_A &= \int_0^t k dt \\ \ln \frac{c_A}{c_{A0}} &= -kt \\ \ln \frac{c_{A0}}{c_A} &= kt \end{aligned} \tag{7}$$

where,

r_A = Reaction Rate for reactant A

k = Rate constant for A

t = Time

c_A = Concentration of A

c_{A0} = Initial concentration of A

Determining concentration as a function of position (l) for a first-order reaction

$$\begin{aligned}\ln \frac{c_{A0}}{c_A} &= k \frac{V}{Q} \\ &= k \left(\frac{(\pi D^2/4) \cdot l}{Q} \right)\end{aligned}$$

$$\theta = \frac{k \cdot \pi D^2/4}{Q}$$

$$\ln \frac{c_{A0}}{c_A} = \theta \cdot l \quad (8)$$

$$\frac{c_{A0}}{c_A} = e^{\theta \cdot l}$$

$$c_A = c_{A0} \cdot e^{-\theta \cdot l}$$

$$c_A = f(l)$$

where,

r_A = Reaction Rate for reactant A, *ppmv/min*

c_A = Concentration of A

c_{A0} = Initial concentration of A

k = Rate constant for A, *1/min*

t = Time, *min*

V = Volume of reactor, m^3 at length l

l = Length in of the reactor, m

Q = Volumetric flowrate, m^3/min

D = Reactor diameter, m

θ = constant, $1/m$

Determining concentration as a function of position (l) for a zeroth-order reaction

$$\begin{aligned}
 c_A &= c_{A0} - kt \\
 &= c_{A0} - k \frac{V}{Q} \\
 &= c_{A0} - k \left(\frac{(\pi D^2/4) \cdot l}{Q} \right) \\
 \theta &= \frac{k \cdot \pi D^2/4}{Q} \tag{9} \\
 c_A &= c_{A0} - \theta \cdot l \\
 c_A &= f(l)
 \end{aligned}$$

where,

r_A = Reaction Rate for reactant A, $ppmv/min$

c_A = Concentration of A

c_{A0} = Initial concentration of A

k = Rate constant for A, $1/min$

t = Time, min

V = Volume of reactor, m^3 at length l

l = Length in of the reactor, m

Q = Volumetric flowrate, m^3/min

D = Reactor diameter, m

θ = constant, $ppmv/m$

C Polymath Code

#Equation to determine hexanal concentration as a function of position on the scaled-up
biofilter

#Constants based on calculations and given tables

$$Q_o = 2831.68 \text{ \#m}^3/\text{min}$$

$$C_{ao} = 2 \text{ \#ppmV}$$

$$k = 2.4259 \text{ \#1/min}$$

$$x = 0.95$$

$$D = 11.41$$

$$\theta = (3.14159 * (D^2)/4)/Q_o \text{ \#1/m}$$

#Differential equations

$$d(C_a)/d(L) = ((-k * \theta) * C_{ao})/(\exp(\theta * k * L))$$

$$C_a(0) = 2$$

$$L(0) = 0$$

$$L(f) = 34.23$$